

- Description

This module manages a smart electric bicycle system, ensuring that:

1. The system continuously monitors the bicycle's current speed (in km/h) and the level of motor assistance (in Watts).
2. If the rider's speed falls below a predefined target speed, the system increases the motor assistance by a fixed percentage to help the rider.
3. If the rider's speed exceeds the target speed, the system reduces the motor assistance by a fixed percentage to conserve energy.

This system helps optimize energy consumption, improve rider comfort, and extend the battery life of the e-bike.

- Specification

module SmartEBikeControl

var currentSpeed, targetSpeed, assistancePower, baseAssistance: real

process updateSpeed(changeAmount: real) returns (currentSpeed: real, assistancePower: real)

functional scenario form:

currentSpeed < targetSpeed && assistancePower = assistancePower * 1.2,
Motor assistance increased by 20%.

||

currentSpeed > targetSpeed && assistancePower = assistancePower * 0.8,
Motor assistance decreased by 20%.

||

currentSpeed = targetSpeed && Motor assistance remains unchanged.

end process

end module

- Java Code

```
package code;
public class c78_SmartEBike {
    private double currentSpeed;
    private double assistancePower;
    private double targetSpeed;

    public c78_SmartEBike(double initialSpeed, double initialAssistance
, double targetSpeed) {
        this.currentSpeed = initialSpeed;
        this.assistancePower = initialAssistance;
        this.targetSpeed = targetSpeed;
    }

    public void updateSpeed(double changeAmount) {
        currentSpeed += changeAmount;
    }
}
```

```

        adjustAssistance();
        System.out.printf("Current Speed: %.2f km/h, Assistance Power: %.2f W%n", currentSpeed, assistancePower);
        if (currentSpeed < targetSpeed) {
            System.out.println("ALERT: Speed below target! Increasing motor assistance.");
        } else if (currentSpeed > targetSpeed) {
            System.out.println("ALERT: Speed above target! Decreasing motor assistance.");
        } else {
            System.out.println("Speed is optimal. Motor assistance remains unchanged.");
        }
    }

    private void adjustAssistance() {
        if (currentSpeed < targetSpeed) {
            assistancePower = assistancePower * 1.2;
        }
        else if (currentSpeed > targetSpeed) {
            assistancePower = assistancePower * 0.8;
        }
    }

    public double getAssistancePower() {
        return assistancePower;
    }

    public double getCurrentSpeed() {
        return currentSpeed;
    }

    public static void main(String[] args) {
        c78_SmartEBike bike = new c78_SmartEBike(15.0, 200.0, 20.0);

        // testcase-VT:
        //     bike.updateSpeed(3.0);
        //     bike.updateSpeed(5.0);
        //     bike.updateSpeed(-3.0);

        // testcase-FT:
        //     bike.updateSpeed(4.580297585357999E290);
        //     bike.updateSpeed(-1.7976931348623157E308);
        //     bike.updateSpeed(1.7976931348623157E308);
    }

```

```

        // testcase-Z3:
//      bike.updateSpeed(3.0);
//      bike.updateSpeed(5.0);
//      bike.updateSpeed(-3.0);

        // testcase-UVT:
        bike.updateSpeed(3.0);
        bike.updateSpeed(5.0);
        bike.updateSpeed(-3.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c78_SmartEBike.java	100.0 %	8	0	8
▼ c78_SmartEBike	100.0 %	8	0	8
main(String[])		0	0	0
c78_SmartEBike(double, double,		0	0	0
adjustAssistance()	100.0 %	4	0	4
updateSpeed(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	105	20,970	21,075
▼ src	0.5 %	105	20,970	21,075
▼ code	0.5 %	105	20,970	21,075
▼ c78_SmartEBike.java	94.6 %	105	6	111
▼ c78_SmartEBike	94.6 %	105	6	111
main(String[])	100.0 %	17	0	17
c78_SmartEBike(double, double,	100.0 %	12	0	12
adjustAssistance()	100.0 %	26	0	26
updateSpeed(double)	100.0 %	50	0	50

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c78_SmartEBike.java	75.0 %	6	2	8
▼ c78_SmartEBike	75.0 %	6	2	8
main(String[])		0	0	0
c78_SmartEBike(double, double,		0	0	0
adjustAssistance()	75.0 %	3	1	4
updateSpeed(double)	75.0 %	3	1	4

5. Fuzz testing instruction coverage: 91.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	102	20,973	21,075
▼ src	0.5 %	102	20,973	21,075
▼ code	0.5 %	102	20,973	21,075
▼ c78_SmartEBike.java	91.9 %	102	9	111
▼ c78_SmartEBike	91.9 %	102	9	111
main(String[])	100.0 %	17	0	17
c78_SmartEBike(double, double,	100.0 %	12	0	12
adjustAssistance()	100.0 %	26	0	26
updateSpeed(double)	94.0 %	47	3	50

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c78_SmartEBike.java	100.0 %	8	0	8
▼ c78_SmartEBike	100.0 %	8	0	8
main(String[])		0	0	0
c78_SmartEBike(double, double,		0	0	0
adjustAssistance()	100.0 %	4	0	4
updateSpeed(double)	100.0 %	4	0	4

7. Z3 instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	105	20,970	21,075
▼ src	0.5 %	105	20,970	21,075
▼ code	0.5 %	105	20,970	21,075
▼ c78_SmartEBike.java	94.6 %	105	6	111
▼ c78_SmartEBike	94.6 %	105	6	111
main(String[])	100.0 %	17	0	17
c78_SmartEBike(double, double,	100.0 %	12	0	12
adjustAssistance()	100.0 %	26	0	26
updateSpeed(double)	100.0 %	50	0	50

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c78_SmartEBike.java	100.0 %	8	0	8
▼ c78_SmartEBike	100.0 %	8	0	8
main(String[])		0	0	0
c78_SmartEBike(double, double,		0	0	0
adjustAssistance()	100.0 %	4	0	4
updateSpeed(double)	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 94.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	105	20,970	21,075
▼ src	0.5 %	105	20,970	21,075
▼ code	0.5 %	105	20,970	21,075
▼ c78_SmartEBike.java	94.6 %	105	6	111
▼ c78_SmartEBike	94.6 %	105	6	111
main(String[])	100.0 %	17	0	17
c78_SmartEBike(double, double,	100.0 %	12	0	12
adjustAssistance()	100.0 %	26	0	26
updateSpeed(double)	100.0 %	50	0	50